



IP

Richer,
More Scalable
Backward Compatible

Ahmed Abdelsalam, Cisco Systems

IP is the Heart of our Digital Economy

IP Best-Effort is > 40-year-old

Impossible to change, isn't it?

Best-Effort for > 40 years

- Destination Address only localize an endpoint
- Development of a myriad of per-domain shim layers to enrich IP
 - MPLS in the core
 - VxLAN in the DC
 - GTP for mobility
- Low scale
 - MPLS cannot summarize => BGP used as an IGP
- Poor End-to-End Services and High Operation Cost
 - Neither TE nor stateless service chaining in VxLAN
 - Translation gateways between domain (VxLAN/MPLS)

IPv6 becomes Programmable

- 128-bit DA becomes:
 - ordered list of uSID instructions (uSID)
 - uSID instruction can bound to anything .. Underlay TE, overlay, service chaining
 - 6 uSID instructions in DA
 - Most deployments requires less than 6 uSID
 - SRH is there for ultra-scale TE
- SRv6 uSID outperforms any IPv4 shim layer: MPLS, VxLAN, GTP

The SRv6 uSID Journey

SRv6 Kick-Off – February 2017 - Rome



SRv6 uSID - Rich Ecosystem

Network Equipment Manufacturers



Merchant Silicon



Access: A100 Edge: K100
Core: Q200, P100
ToR/Spine: G100, G200

DNX – Jericho
XGS – Tomahawk

Open-Source Applications



Open-Source Networking Stacks



Smart NIC / DPU



Partners



SRv6 uSID - Fully Standardized

- SR Architecture – RFC8402
- IPv6 SR header – RFC 8754
- Network Programming – RFC 8986
- **SRv6 uSID – RFC9800**
- BGP Services – RFC 9252
- ISIS - RFC9352
- OAM – RFC 9259
- Policy Architecture – RFC 9256
- PCEP – RFC 9603
- BGP-LS – RFC 9514
- Flex Algo – RFC 9350
- Performance Management – RFC 9503
- SRv6 SID Block – RFC 9602
- **SR Policy Provisioning via BGP [\[link\]](#) – RFC XXXX**
- **SR Policy Advertisement in BGP-LS [\[link\]](#) – RFC XXXX**
- **BGP Color-Aware Routing [\[link\]](#) – RFC XXXX**

Across the portfolio - Skin in the Game

- 2016: uSID idea is found
- 2019: uSID productization is triggered across Cisco Portfolio
- 2021: first uSID deployments
- 2025: uSID is Proposed Standard, all vendors have joined uSID
 - 2019 plan was proved correct and no implementation change was required!
- >85k Cisco routers deployed

AI DC – Backend

SRv6 uSID for AI @ Microsoft

“In the AI backend network, the training traffic is very bursty and each of the flow is elephant flow.

Traditional ECMP does not work for the AI backend network and it's particularly bad because the training job is highly synchronized. If flows collide on a link and it gets congested - it drags the whole job and delay the whole job significantly.

We use source routing for the backend network

SRv6 is a very perfect technology to do the source routing in the AI backend network.”

Guohan Lu, AI Fabric, Microsoft. April 2025.

[OCP Spring 2025 - Public Presentation Recording](#)



SRv6 uSID for AI @ Microsoft

“We Like SRv6 for several reasons:

*The first, the source routing capability, this is particularly useful for our users who want precise control over the traffic flows for **optimal traffic performance**.*

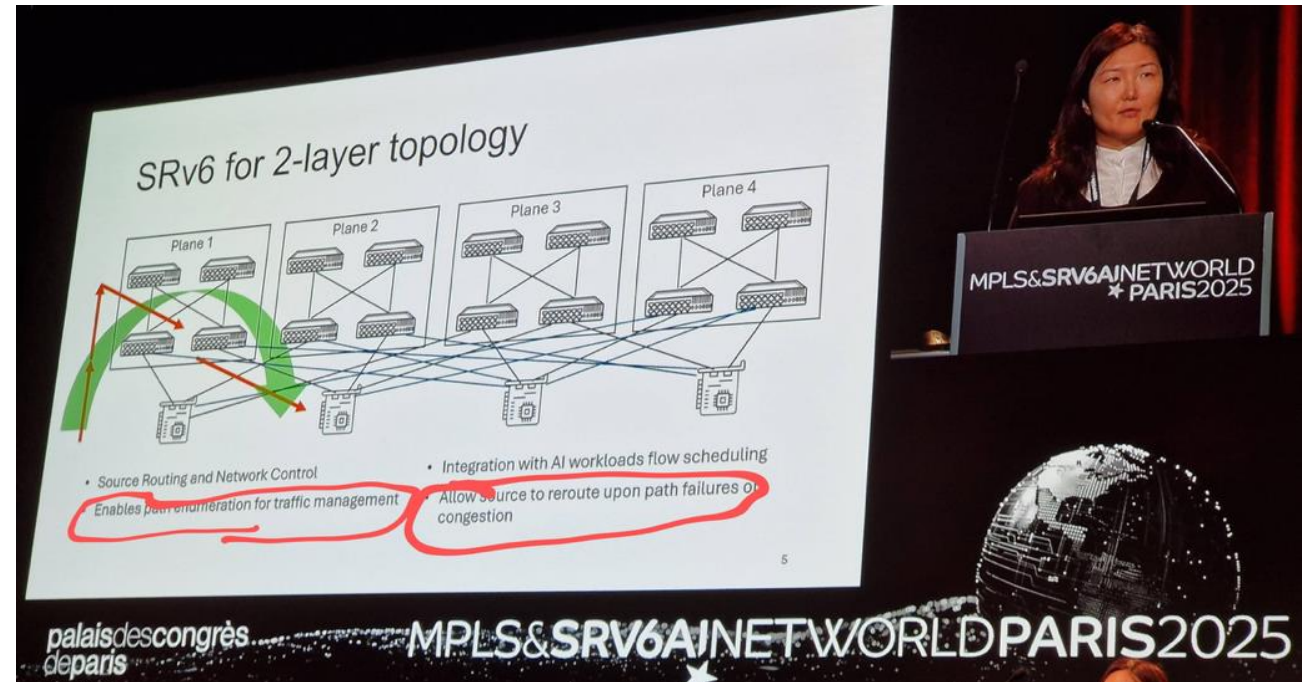
The second, SRv6 enable detailed path enumeration allowing to manage the traffic for low latency requirements

*The third, it enables **the right coupling between AI workload flows and the scheduler**.*

*Finally, **the network failure detection and response is at the source side. The source can recalculate the path and re-route the traffic without the dependency on the traditional routing protocols.**”*

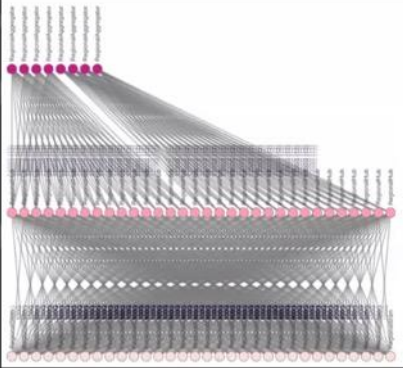
Rita Hui, MSFT AI Fabric, March 2025

[SRv6 World - Public Presentation Recording](#)



SRv6 uSID for AI @ Microsoft

Networking for AI



- Extreme scaling
800G → 1.6T → 12.8T
- Low latency
<2ms <1ms
- High bandwidth
>12.8 Tbps
- Leading open source
SONiC

- High **performance & security** network for AI (for all types of accelerators - Nvidia, AMD, Maia)
- Dedicated, **low latency** WAN for AI
- First large-scale co-operative transport (host + network) for **fast failover and load balance** via SRv6 on SONiC
- Millisecond-level, **high-frequency telemetry** for troubleshooting
- BareMetal host with **SDN policies**
- Secure and high-bandwidth connectivity via **Private Link** to storage for training



- Deepak Bansal, Corporate VP, MSFT @ [Hot Interconnect 2025 - Public Presentation](#)

SRv6 uSID for AI Backed



Workgroup: SRv6 Operations
Published: 18 August 2025
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J. Tantsura
NVIDIA
K. Patel
Arrcus, Inc.

SRv6 for Deterministic Path Placement in AI Backends

Abstract

This document describes the use of SRv6 to enable deterministic path placement in AI backends, optimizing load balancing and congestion control for predictable GPU workloads.

Cisco, Oracle Cloud, IBM, NVIDIA, Arrcus

Benefits

- * **Deterministic Path Placement:** SRv6 allows the NIC to control the path of each flow through the fabric.
- * **Minimum-MTU:** A plain outer IPv6 encapsulation allows to encode 6 uSIDs in the outer DA. This implies that without the need of additional extension headers, only with 40Bytes of IPv6 encapsulation, we can encode up to 6 intermediate waypoints allowing to enforce a path in a 3-tier Clos network. This is sufficient to control a path hop-by-hop (link by link) through a leaf, spine, super-spine, spine, leaf.
- * **Congestion Feedback Loop:** Instant rerouting at the source based on ECN, in-band measured One-Way and Two-Way latency, Packet Trimming feedback and in-band packet loss, without any dependency of routing protocols. There is neither any control-plane signaling involved between the GPU and the fabric, nor between the AI orchestrator and the fabric devices.
- * **Standardization:** Open, vendor-agnostic implementation
- * **Ease of operation:** as opposed to black-box proprietary solution which packs opaque layer-2 optimization, the SRv6 solution is minimalistic, IP based, fully standardized and a rich ecosystem (vendor, merchant and open source). The deterministic and open nature of the solution simplifies troubleshooting.

<https://datatracker.ietf.org/doc/draft-filsfils-srv6ops-srv6-ai-backend/>

SRv6 uSID in SONiC (Mature, Deployed, Rich Ecosystem)

- **Mature:**

- AI Backend: Static uSID Fabric - SDN controller
- L3VPN: IPv4/IPv6 - Static & BGP (RFC9252)
- SRv6 GRT: IPv4/IPv6 - Static & BGP (RFC9252)
- Underlay Traffic Engineering: Static & IS-IS (RFC9352)
- Static steering of IPv4/IPv6 traffic over SRv6 uSID
- SRv6 SID Manager - F3216/GIB/LIB/WLIB

- **Deployed:**

- Alibaba – In production across all China
- Microsoft – AI backend

- **Rich Ecosystem:**

- Mainline: 350+ PR and 1600+ commits across SAI/SONiC/FRR.
- Contributors: Cisco, Microsoft, Alibaba, Broadcom, Nvidia



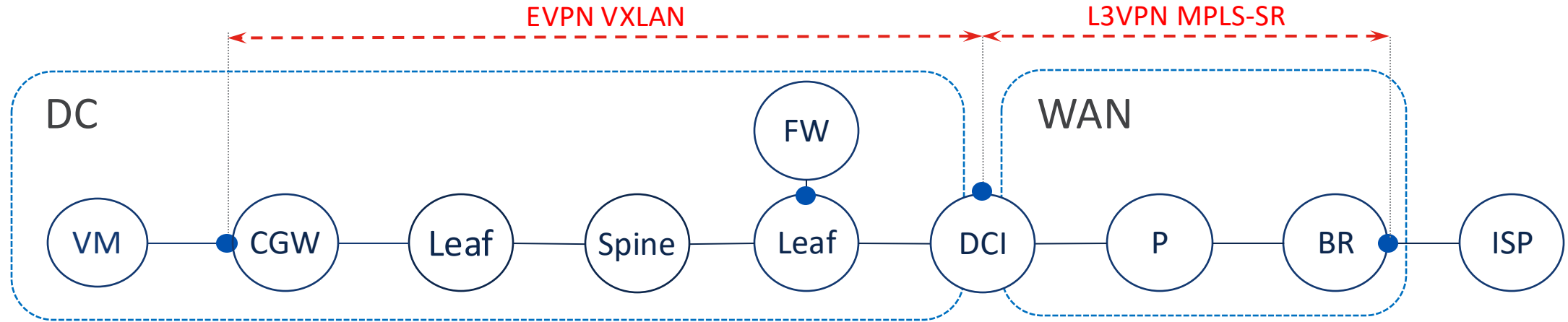
Alibaba eCore - across China
Eddie Ruan - Paris2024



Microsoft - AI Backend
Rita Hui - Paris2025

AI DC – Frontend

Legacy architecture of Frontend DC

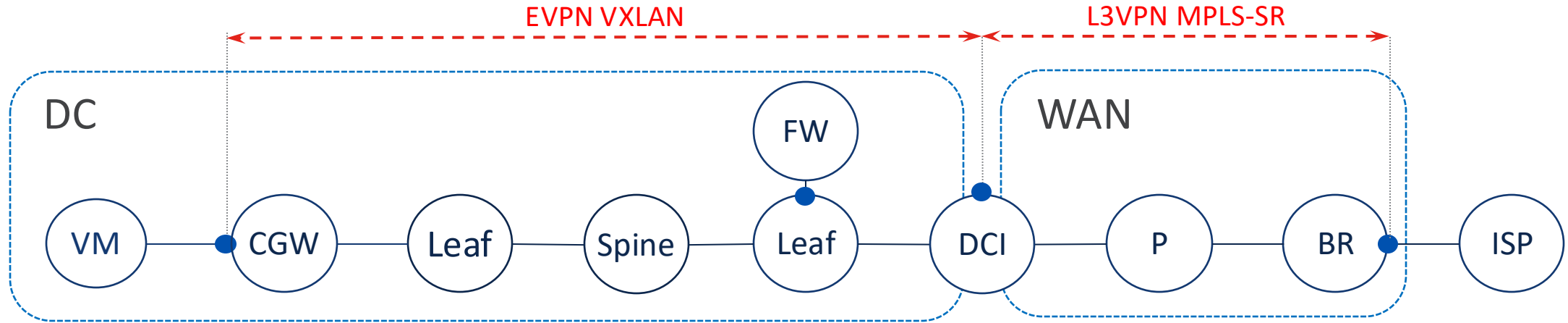


- IPv6 only infrastructure in the Data Center
- CGW (Cloud Gateway) and FW are NFV's running on hosts. Nebius develops them
- VXLAN based overlay between CGW and DCI
- DCI does “stitching” between EVPN VXLAN and L3VPN MPLS-SR

Alexey Gorovoy, Nebius, March 2025
[SRv6 World - Public Presentation Recording](#)

Legacy architecture of Frontend DC

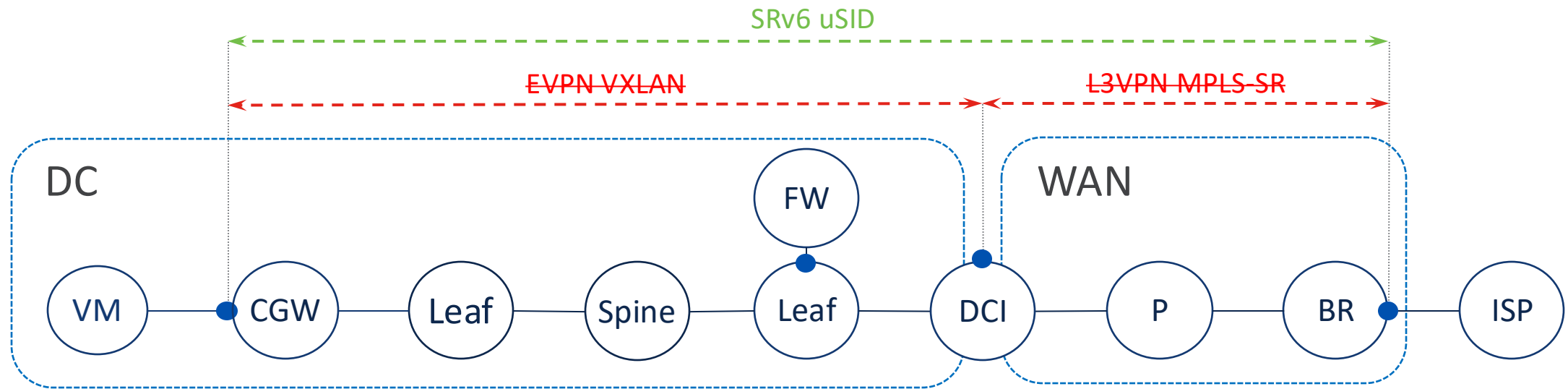
Limitation



- VXLAN
 - No traffic engineering capabilities inside the Data Center
 - Service chaining with VXLAN requires specific routing design (PBR, Default GW, VRF/VLAN hand-off, etc.)
 - Majority vendor implementations of VXLAN still require IPv4 loopbacks in the Underlay
- SR-MPLS
 - Lack native Data Center optimisations and not applicable in the DC domain
- Requires "stitching" gateway functionality at the DCI

Alexey Gorovoy, Nebius, March 2025
[SRv6 World - Public Presentation Recording](#)

SRv6 in the Front-End DC



- SRv6 uSID

- Unified End-to-End design from the DC Front End, through the metro, up to the Peering
- Replaces the legacy design with VXLAN in the DC and SR-MPLS in the metro network.
- Firewall Service insertion – stateless
- Native TE

Alexey Gorovoy, Nebius, March 2025
[SRv6 World - Public Presentation Recording](#)

AI DCI

DCI is crucial for AI

- Workload spans clusters scattered over a regional network

AI WAN - MSFT



Satya Nadella • Following
Chairman and CEO at Microsoft
8h •

If intelligence is the log of compute... it starts with a lot of compute! And that's why we're scaling our GPU fleet faster than anyone else.

Just last year, we added over 2 gigawatts of new capacity – roughly the output of 2 nuclear power plants.

And today we're going further, announcing the world's most powerful AI datacenter, located in southeastern Wisconsin.

Fairwater is a seamless cluster of hundreds of thousands of NVIDIA GB200s, connected by enough fiber to circle the Earth 4.5 times.



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Inside the world's most powerful AI datacenter

Sep 18, 2025 | [Scott Guthrie - Executive Vice President, Cloud + AI](#)

AI WAN: Connecting multiple datacenters for an even larger AI supercomputer

These new AI datacenters are part of a global network of Azure AI datacenters, interconnected via our Wide Area Network (WAN). This isn't just about one building, it's about a distributed, resilient and scalable system that operates as a single, powerful AI machine. Our AI WAN is built with growth capabilities in AI-native bandwidth scales to enable large-scale distributed training across multiple, geographically diverse Azure regions, thus allowing customers to harness the power of a giant AI supercomputer.

This is a fundamental shift in how we think about AI supercomputers. Instead of being limited by the walls of a single facility, we're building a distributed system where compute, storage and networking resources are seamlessly pooled and orchestrated across datacenter regions. This means greater resiliency, scalability and flexibility for customers.

TE is a MUST for AI DCI

- The topology is less symmetric
- The capacity is more scarce
- Latency matters more
- Weirdness:
 - Shortcuts between clusters only for specific workloads
 - SRLG's

SRv6 TE is the Perfect Fit for AI DCI

- Perfect fit for the “IP nature” of legacy DCIs
 - In fact, most use pre-SRv6 IPinIPinIP source routing
- Perfect fit for the SONiC expansion in AI DCI
 - Leverage the richness/maturity from AI DC
 - SONiC or Custom Cloud OS
- Perfect fit for HW
 - SRv6 is rich/mature across all merchant silicon (e.g., Cisco S1)
- Perfect End-to-End model
 - SRv6 from DC to DC through DCI

MPLS TE does NOT fit AI DCI

- No MPLS Legacy in DCIs
 - MPLS requires pervasive deployment
- No MPLS in SONiC and Custom OS
 - too much investment, too slow, too late
- No MPLS on DC HW
 - too much testing, significant gaps expected
- Breaks the End-to-End model
 - SRv6 to MPLS and then MPLS to SRv6 at each DC border is too complex & costly

AI DCI Interest

- Much interest through the AI/Cloud industry
- Easy to understand
 - AI => TE
 - DCI => IP
 - IP + TE = SRv6
 - SRv6 Rich and Mature in SONiC (Alibaba Ultra Scale)

Ultra-Scale Alibaba DCI Deployment

- Watch the OCP session
- Read the upcoming detailed design guide
- Ultra Scale DCI deployment across China
- Cisco S1/SONiC/SRv6 based
- Alibaba / Cisco partnership since 2021

Thu, October 16, 10:45am - 11:05am | SJCC - Concourse Level - 220B

eCore: Architecture and Whitebox Evolution of Alibaba's Service-oriented DCI network

● Networking

eCore is Alibaba's cutting-edge Data Center Interconnect (DCI) network, engineered to redefine scalability and efficiency through a modular, whitebox-based architecture and service-aware traffic forwarding powered by SRv6. This talk, presented jointly by Alibaba and Cisco, delves into its design philosophy, deployment challenges, and measurable impact—alongside Alibaba's custom whitebox solutions and roadmap for the future.

Roy Jiang - Network Architect at Alibaba Cloud

OCP Global Summit, San Jose, Oct 2025

AI WAN

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AI WAN - Alibaba Deployment

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More SRv6 uSID deployments

20 MPLS networks into 1 SRv6 network



IPv6 Addressing & Summarization

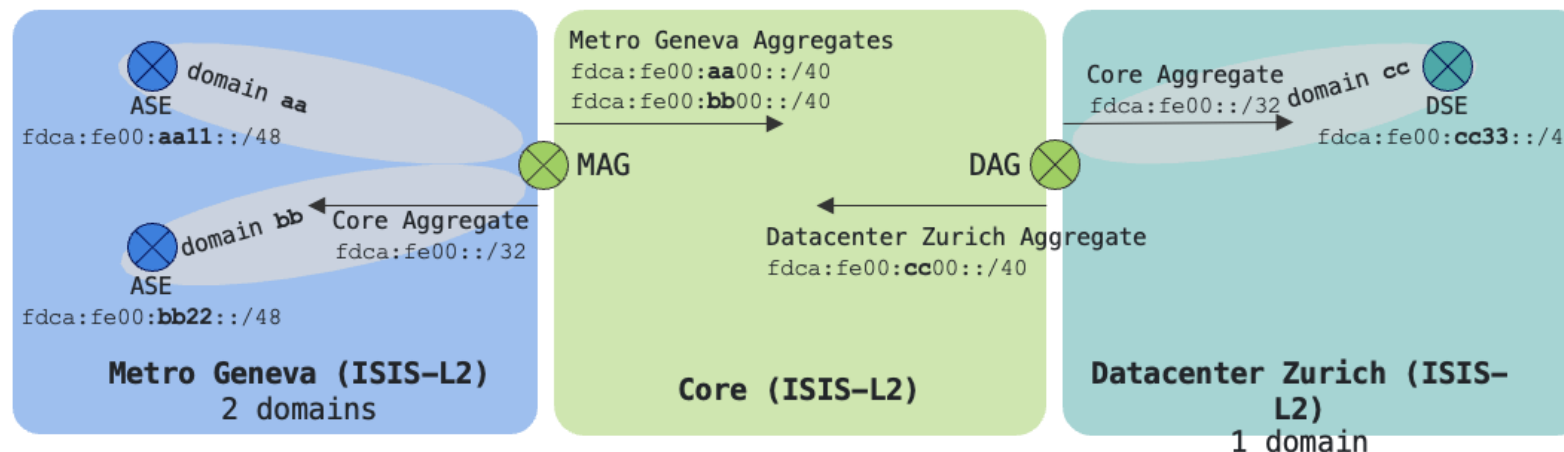
The beauty of SRv6 ! Keeping the IGP tables small and clean.

Addressing concept

uSID Locator	<code>fdca:fe0G:DDNN::/48</code>	Loopback0	<code>fdca:fe00:DDNN::1/128</code>
Domain Summary		<code>fdca:fe0G:DD00::/40</code>	

Summarization example (Algo 0 only)

- in core $n \times /40$ domain summaries, min 1 from each metro and datacenter
- in metro/datacenters $1 \times /32$ to reach any other metro/datacenter



Rakuten Mobile SRv6 uSID IP Transport Transformation



R

47
Prefectures

59
IGP domains

15000
SRv6 nodes

Largest
u-SID SRv6
deployment

ISIS, BFD, BGP
(vpn4, vpnv6, rt-filter,
evpn, bgp-ls)

4G,5G, L2VPN
(EVPN), L3VPN

SRv6 Deployment Outcomes

98%

Reduction in ISIS LSP in Core

73%

Reduction in total ISIS routes in
Core

96%

Reduction in number of BGP
sessions on Core routers
55 to 2

SRv6 transforms
Rakuten IP Transport
network to world
class converged 5G
SDN-ready network

90%

Reduction in total routes to data
centers
6500+ → 500

86%

Reduction in config touch-points
13-15 → 2

90%

Improvement in convergence
time
500ms to 50ms

94% reduction in unique sites flap post SRv6 migration

R

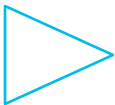
From a complex infrastructure to simplicity with SRv6



Rijkswaterstaat
Ministry of Infrastructure
and Water Management



Critical network infrastructure to support road and maritime traffic and water management



...and SRv6 powers the Outernet!

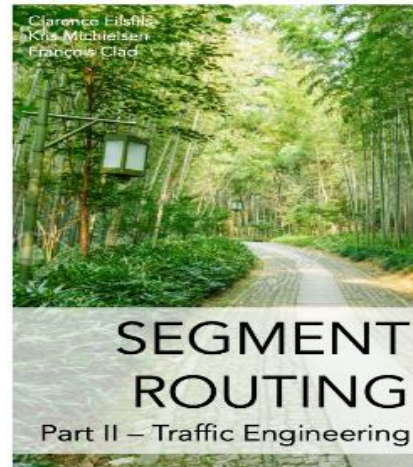
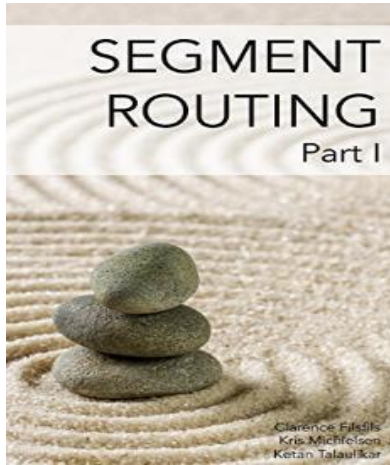


Rivada & Cisco Creating Blueprint for
Advanced Data Networking from Space.

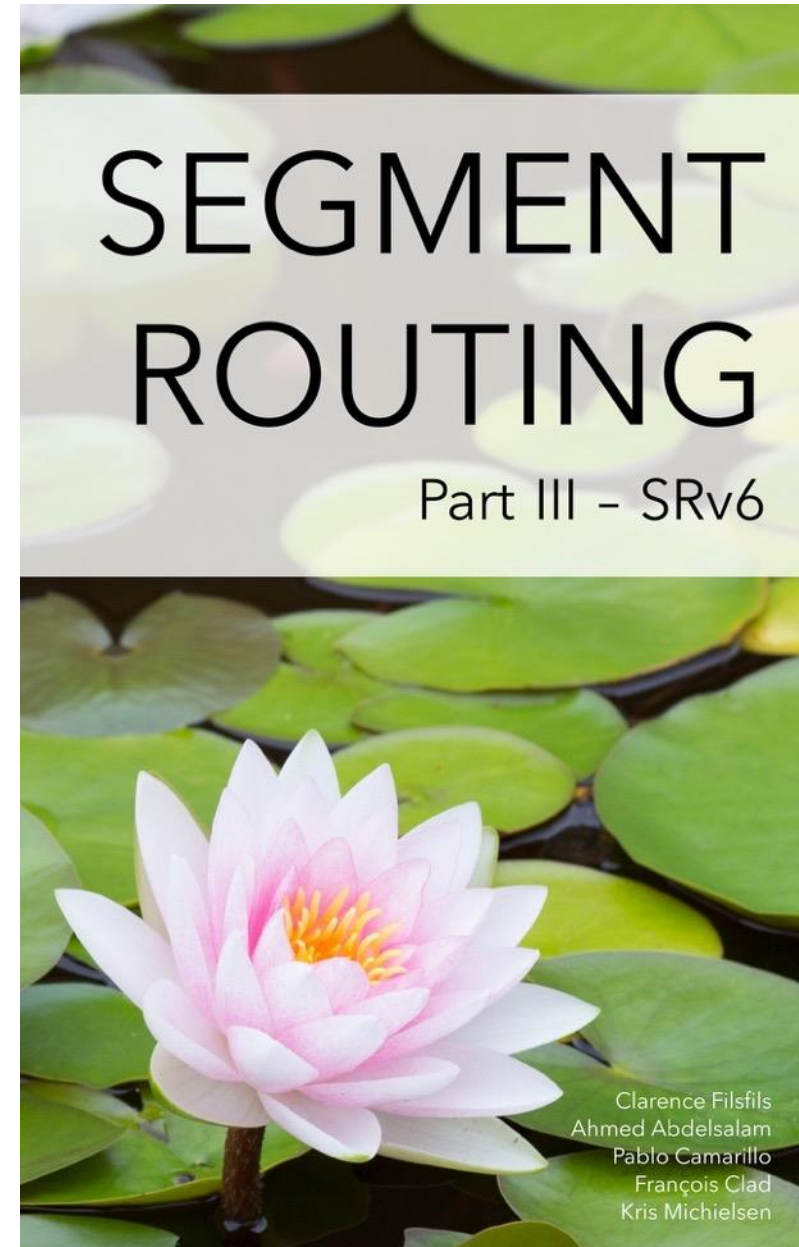
[LINK](#)

In collaboration with Cisco, Rivada is ready to [converge the Outernet with traditional terrestrial networks](#). Cisco provides high performance routing using IOSXR based NCS540, NCS55/5700 and 8000 product line, to build agile service network (ASN) with rich [SRv6](#)/EVPN feature set, advanced QoS, robust Security and AI enabled automation. Built with standardized API, the solution will be able to provide [end to end network slicing and service assurance across non-terrestrial and terrestrial networks](#).

Stay up-to-date



segment-routing.net



<https://www.amazon.com/dp/B0D6GWWRWH>

Simplicity Always Prevails



